Working title: What Changes in the High-RL Regime? A From-Scratch Testbed

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1 Introduction

Talk about:
- Frontier RL has grown from a thin layer to roughly pretraining scale. Graded per-model estimates of RL / pretraining compute: 0.01–0.1 in 2024 (grade C), 0.3–1 in H2 2025 (D), about 1–2 in 2026 (D, mixed units). No lab publishes the FLOP ratio. Closest statements: xAI's Grok 4 ran RL "at pretraining scale"; DeepSeek-V3.2's post-training budget exceeds 10% of pretraining cost. (REVIEW.md Q3; quote the range with its units and grade, never as one number.)
- Most of what we believe about what RL does was measured in the small-RL regime: Yue et al.'s RL is a median 1.2×10^{-4} of pretraining FLOPs, and no create-vs-elicit test on a pretrained LLM with a computable ratio exceeds about 2×10^{-2} (REVIEW.md bottom line 1–2).
- Possible opener: "Reinforcement learning is no longer a thin layer on top of pretraining."
- Cite: [1, 2]. Not in references.bib yet: DeepSeek-V3.2 (arXiv 2512.02556), the Grok 4 post, the Epoch estimates.

[TODO: write P1]

Talk about:
- The low-RL picture: post-training mostly selects among behaviours (personas) the model learned in pretraining, the persona selection model (PSM). [TODO: PSM citation: ask Dmitry which source]
- Misalignment risks crop up under heavy RL: reward hacking, and strategies that never appeared in pretraining. [TODO: citations, none in references.bib yet: e.g. MacDiarmid et al. 2025 (misalignment emerging from reward hacking in production RL), Baker et al. 2025 (monitoring CoT, obfuscation under pressure)]
- End on the three questions from the notes: (1) at what point does the PSM break down? (2) how much RL does it take to remove a behaviour instilled in pretraining, or to install one? (3) are there proxy metrics that tell you that you are in this regime?

[TODO: write P2]

Talk about:
- At frontier scale, high-RL experiments are expensive and the pretraining data is uncontrolled: you cannot knock a skill out of the corpus and then ask whether RL builds it.
- Our setting: a small transformer trained from scratch on natural-deduction proofs, rewarded by a Lean-checked verifier, so every pretraining token is ours. [TODO: parameter count of the GRPO base (S1g3); the EI runs' ALiBiGPT 6×384 has 9,560,832 (REVIEW.md Q5)]
- Dmitry (Oct 22 sync): toy models are how we reach the "high RL, low pretraining" regime cheaply.
- Our EI runs spend 0.9–19× pretraining FLOPs on RL (RL-only count; 9–180× by GPU-time): at or above the frontier band, about 4–5 orders of magnitude above Yue et al.'s median (REVIEW.md Q5). [TODO: the same ratio for the GRPO runs (merged recipe + KL 0); not computed yet]
- Nearest from-scratch organisms: Interplay (Zhang et al., arXiv 2512.07783; RL $\approx 0.4\times$ pretraining) and Echo Chamber (Zhao et al., arXiv 2504.07912). Dan's knockout is the only controlled create-vs-elicit test above $1\times$ (REVIEW.md implication 1).

[TODO: write P3]

Talk about:
1. We are in a high-RL regime, and not just by GPU-hours: RL / pretraining FLOPs, capabilities that emerge only during RL, performance on the held-out textbook set (never trained on). (§2, Leon + Dan; 1–2 pages)
2. Proxy metrics for "the model has diverged from pretraining" that also apply to real LLMs: EDL (the best Dmitry has seen), KL to the pretrained model (Robbie's pick). Do they move smoothly or jump? (§3, Charles + Robbie)
3. The metrics line up with behaviour: the model reward hacks more, and it uses proof strategies pretraining never showed it (Dan's proof-strategies taxonomy; the reward-hacking figures in `code/experiments/current/reward_hack_inoculation_20261023/`). (§4)
4. What this predicts for language models, and the experiments to run there. (§5)

Preprint.

[TODO: write P4]

References

[1] Daya Guo et al. "DeepSeek-R1 Incentivizes Reasoning in LLMs through Reinforcement Learning". In: *Nature* 645.8081 (Sept. 2025), pp. 633–638. DOI: 10.1038/s41586-025-09422-z.

[2] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. *Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?* Nov. 24, 2025. DOI: 10.48550/arXiv.2504.13837. arXiv: 2504.13837 [cs.AI]. URL: http://arxiv.org/abs/2504.13837 (visited on 06/15/2026). Pre-published.

Figure 1 (teaser), to choose.
(a) RL / pretraining compute on a log axis: the create-vs-elicit papers, the graded frontier band, our runs.
Dan has this plot (origin/dan, rl-compute-share `figures/ratio.png`); redraw in `style.py` and add the GRPO runs.
(b) Cartoon: low RL selects behaviour learned in pretraining; high RL builds new behaviour.

Figure 1: [TODO: caption]